

Six current first-use routes. The HTML article carries links, examples, and full scope boundaries.

1. Prepare and specify

Put responses into long or wide data and state the covariance question.

Functions: `traits()`,
`gllvmTMBcontrol()`, `latent()`, `indep()`,
`dep()`

Start with ordinary covariance; add a relationship source only when the study design supplies one.

2. Fit

Fit every stacked-trait model through one entry point.

Functions: `gllvmTMB()`

Use a supported response family and retain the formula that defines the estimand.

3. Check fit health

Check convergence, gradients, Hessian information, and fitted-response mismatch before interpretation.

Functions: `check_gllvmTMB()`,
`gllvmTMB_diagnose()`,
`predictive_check()`

A syntactically valid model can still be weakly identified for its data.

4. Interpret covariance

Extract rotation-invariant covariance, correlation, and shared-variance summaries first.

Functions: `extract_Sigma()`,
`extract_correlations()`,
`extract_communality()`,
`extract_ordination()`

Loadings and ordination scores are orientation-dependent; interpret them after covariance summaries.

5. Choose a target-specific uncertainty route

Inspect available methods and their returned status rather than assuming a generic confidence interval.

Functions: `profile_targets()`,
`confint_inspect()`, `bootstrap_Sigma()`,
`loading_ci()`

Interval availability and calibration depend on the estimand, family, and covariance tier.

6. Report, predict, or simulate

Make report-ready tables and figures, predict fitted responses, or simulate new responses.

Functions: `extract_Sigma_table()`,
`plot_Sigma_heatmap()`,
`plot_correlations()`, `predict()`,
`simulate()`

Name the target and scale in every reported table or figure.